Probability And Statistics

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Lecture 1: Probability and Counting

Sample Space S of an experiment is the set of all possible outcomes of the experiment. An **event** A is a subset of sample space S, we say A **occured** if the actual outcome of the experiment is in A.

$$P(A) = \frac{|A|}{|S|}$$

The probability of something happening is the events satisfying that outcome over the total number of outcomes. Formulas that follow from this include $P(\varnothing) = 0$ and $P(S) = 1$.

Sampling table: choose k objects out of n

	order matters	**order doesn't matter**
with replacement	n^k	$\binom{n+k-1}{k}$
without replacement	$\frac{n!}{k!}$	$\binom{n}{k}$

Further formulas:

$$P(A^c) = 1 - P(A)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cup B) = P(A) + P(B)$$

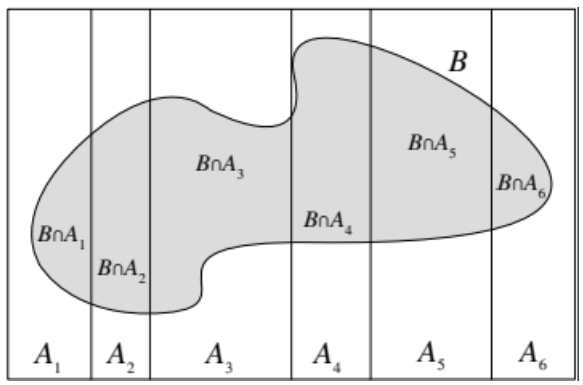
Lecture 2: Conditional Probability

$P(A|B)$ is the **posterior probability** of A. Assuming $P(B) > 0$, $P(A|B)$ is the probability of A given the evidence B. $A|B$ **is not an event!!!**

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B|A)P(A)}{P(B)}$$

The Law of Total Probability:

$$P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$$



Lecture 3: Random Variables

A **Random Variable** maps the sample space onto the real number line. The **Probability Mass Function (PMF)** is a function that describes the distribution of a random variable X. This is formally given by:

$$P(X = x) = \{s \in S : X(s) = x\}$$

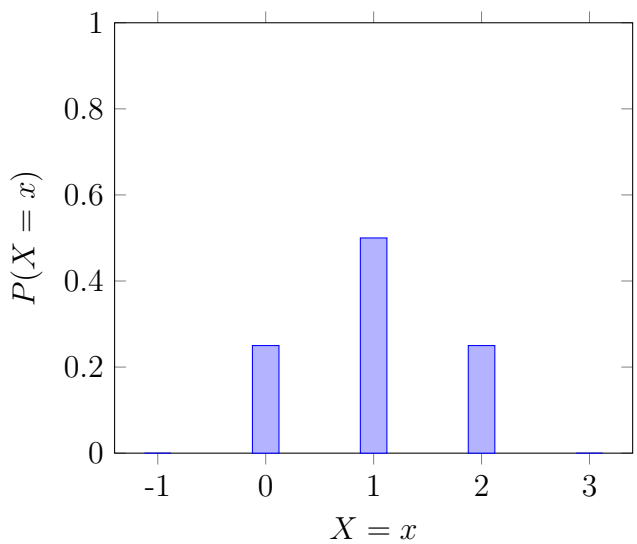
Here X is the random variable, S is the sample space, and x is a real number.

Example: Consider an experiment where we toss a coin twice. The sample space is $S = \{HH, HT, TH, TT\}$. Let random variable X be the number of heads. This defines random variable X as:

$$X(TT) = 0, X(TH) = X(HT) = 1, X(HH) = 2$$

The PMF is defined as:

$$P(X = 0) = \frac{1}{4}, P(X = 1) = \frac{1}{2}, P(X = 2) = \frac{1}{4}$$



Random variable: maps from sample space S to real number $\mathbb{R}$
PMF: maps real number $\mathbb{R}$ to probability $[0, 1]$

A PMF is *valid* if:

- it is nonnegative (no negative probabilities)
- it sums to 1

<u>Named Distributions:</u>

$$X \sim \text{Bern}(p)$$

Bernoulli Distribution: X has a Bernoulli distribution if there are only two outcomes possible, with the probability for success being p. This means $P(X = 1) = p$ and $P(X = 0) = 1 - p$.

$$X \sim \text{Bin}(n, p)$$

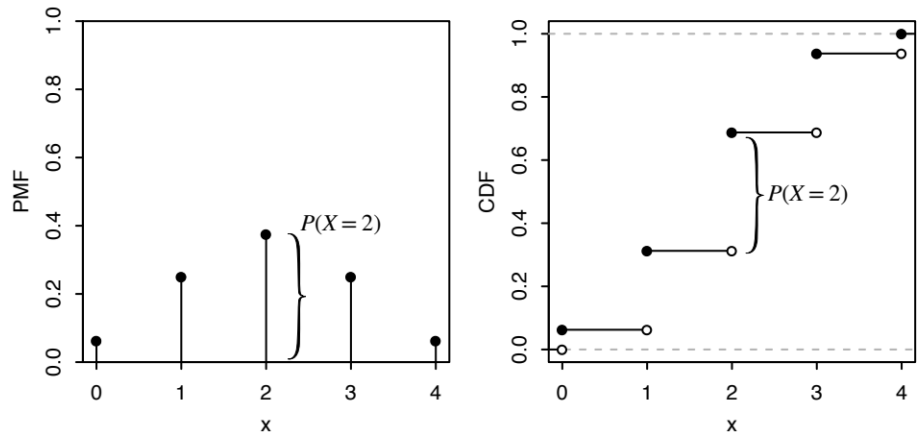
Binomial Distribution: X has a binomial distribution if n independent Bernoulli trails with success probability p are performed. Here X represents the number of successes. The PMF for a binomial distribution equals:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$X \sim \text{DUnif}(C)$$

Discrete Uniform Distribution: X has a discrete uniform distribution if X has equal possibility to be any real number defined in set C. The PMF for a discrete uniform distribution equals $P(X = x) = \frac{1}{|C|}$ for $x \in C$ (else 0).

The **Cumulative Distribution Function (CDF)** of a random variable X is given by the function $F_X(x) = P(X \leq x)$.



Any CDF has the following properties:

- It is increasing
- It is right-continuous
- $\lim_{x \to -\infty} F_X(x) = 0$
- $\lim_{x \to \infty} F_X(x) = 1$

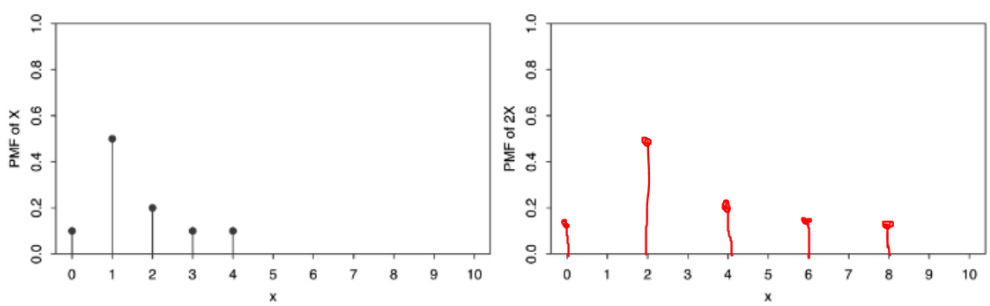


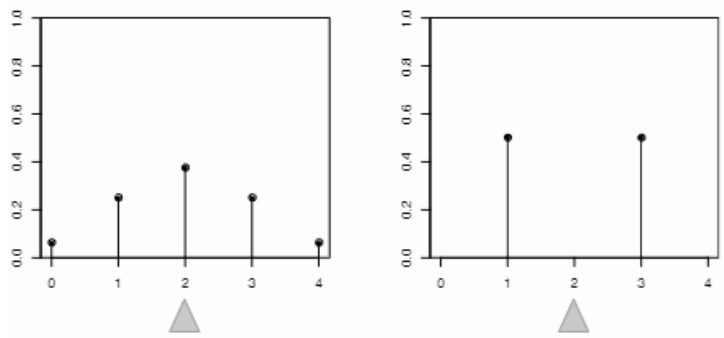
Figure 1: Multiplication of Random Variables

Lecture 4: Expectation

The **Expected Value** (also called expectation or mean) of a random variable X is defined as:

$$E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i)$$

Here $P(X = x_i)$ is refered to as the **weight** of that value of x. If random variables X and Y have the same distribution then they also have the same expectation, but two variables don't *need* to have the same distribution to have the same expectation:



For any two random variables X and Y, and any constant c:

$$E(X) + E(Y) = E(X + Y)$$

$$E(cX) = c \cdot E(X)$$

Law of the Unconscious Statestician:

$$E(g(X)) = \sum_{x} g(x) P(X = x)$$

Example: Let random variable X be defined for $0, 1, 2, ...$ with probabilities $p_0, p_1, p_2, ...$ so its PDF equals $P(X = n) = p_n$. Then X^3 is defined for $0^3, 1^3, 2^3, ...$ with probabilities $p_0, p_1, p_2, ...$ (see figure 1). This gives:

$$E(X) = \sum_{n=0}^{\infty} n \cdot p_n$$

$$E(X^3) = \sum_{n=0}^{\infty} n^3 \cdot p_n$$

Variance:

$$\mathrm{Var}(X) = E(X^2) - E(X)^2$$

Standard Deviation:

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$$

Poisson Distribution:

$$X \sim \mathrm{Pois}(\lambda)$$

X has a Poisson distribution defined by a parameter λ. If X has a Poisson distribution then $E(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$. Its PMF is:

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Let $A_1, ..., A_n$ be events with $P(A_i) = p_i$. You can use a Poisson distribution if n is large and p_i is small. The λ for this distribution is the average number of occurences (expected value) $\lambda = n \cdot p_i$.

Lecture 5: Continuous Random Variables

A random variable has a continuous distribution if its CDF is differentiable. The CDF can be undifferentiable at various (end)points, as long as it is differentiable everywhere else. The PMF of a continuous random variable is called a **probability density function (PDF)**.

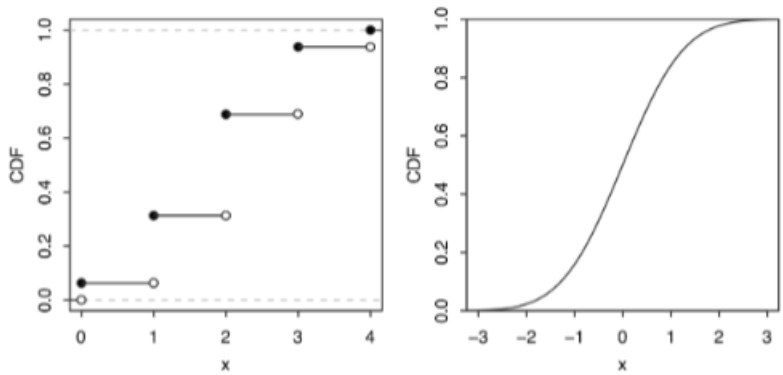
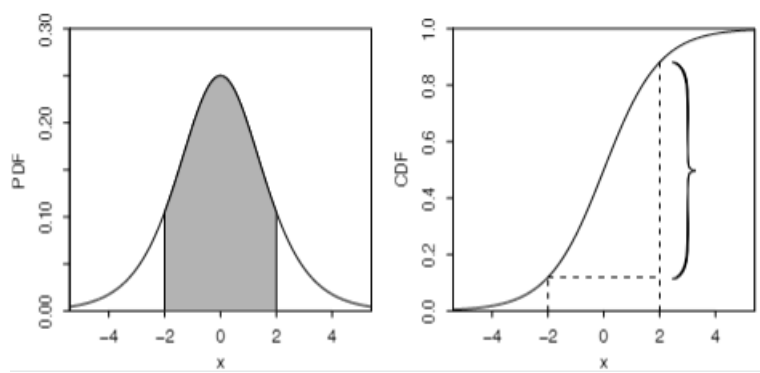


Figure 2: CDF with discrete r. v. (left) and continuous r. v. (right)

For a continuous random variable X, the *PDF* of X is the derivative of the *CDF* of X. Vice versa, the *CDF* of X is the integral of the *PDF* of X.



For a continuous random variable X, $P(a < X < b)$ equals the area under the graph of the PDF of X. Thus, $P(a < X < b) = \int_a^b f_X(x)\,dx$ where $f_X(x)$ is the PDF of X.

To calculate the **expected value** of a continuous random variable, use the same formulas as a discrete random variable, but use an integral ($\int_{-\infty}^{\infty}$) instead of a sum. For a PDF of a continuous random variable to be valid, it must be:

- nonnegative
- integrate to 1 ($\int_{-\infty}^{\infty} f(x)\,dx = 1$)

Uniform Distribution:

$$X \sim \text{Unif}(a, b)$$

X has a uniform distribution if its PDF is $\frac{1}{b-a}$ if $a < x < b$, else 0.

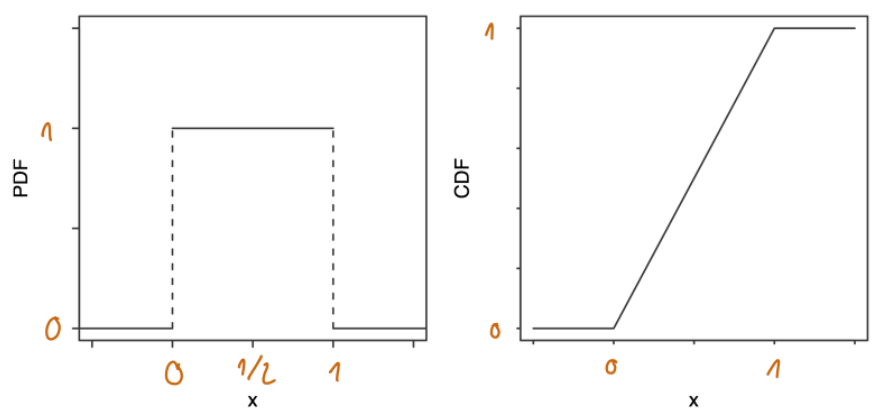


Figure 3: $X \sim \text{Unif}(0, 1)$

Standard Normal Distribution:

$$Z \sim N(0, 1)$$

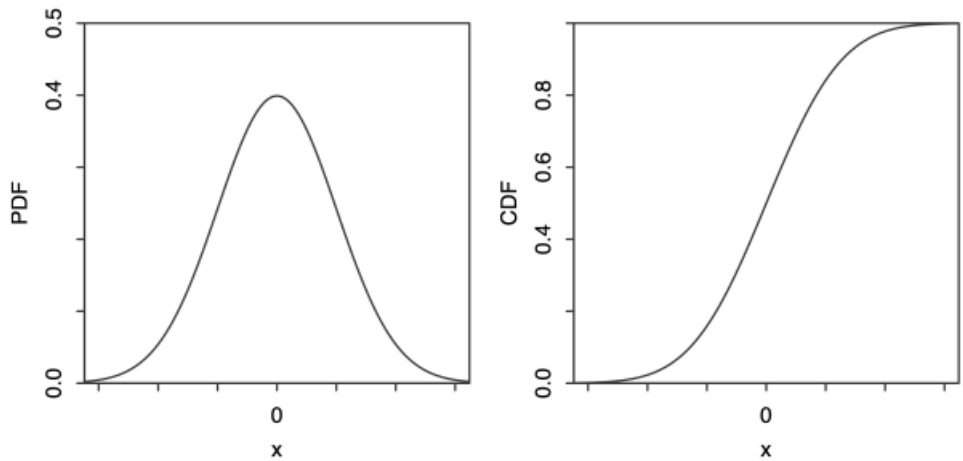
A continuous random variable Z is said to have the standard normal distribution if its PDF (φ) is given by:

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

for $-\infty < z < \infty$.

The standard normal CDF (ϕ) is the area under the standard normal PDF (φ):

$$\phi(z) = \int_{-\infty}^{z} \varphi(t)\, dt = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\, dt$$



Notice the symmetry of the PDF around 0. This means Z has an expected value of $E(Z) = 0$ and a variance of $\mathrm{Var}(Z) = 1$.

If $Z \sim N(0, 1)$, then $X = \mu + \sigma Z$ is said to have a normal distribution with mean μ, variance σ^2 and standard deviation σ, for any real μ and $\sigma > 0$. This is denoted as:

$$X \sim N(\mu, \sigma^2)$$

Lecture 6: Joint Distributions

The **joint PMF** of discrete random variables X and Y is defined as $P(X = x, Y = y)$. The **marginal PMF** of discrete random variables X and Y equals:

$$P(X = x) = \sum_y P(X = x, Y = y)$$

The **conditional PMF** of discrete random variable Y given $X = x$ is:

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

If random variables X and Y are **independent** and **discrete**, then:

$$P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$$

Example: Suppose we have researched a small portion of the population on the effects of smoking. Random variables X and Y are both Bernoulli distributed. $X = 0$ represents a non-smoker, $X = 1$ represents a smoker. $Y = 0$ means they are not developing cancer, $Y = 1$ means they are developing cancer. For this research $n = 100$.

	Y = 0 (no cancer)	Y = 1 (cancer)
X = 0 (non smoker)	72	3
X = 1 (smoker)	20	5

Some examples:

- $P(X = 0, Y = 0) = \frac{72}{100}$ (non smoker has no cancer)
- $P(X = 1, Y = 1) = \frac{5}{100}$ (smoker has cancer)
- $P(X = 1) = \frac{20}{100} + \frac{5}{100} = \frac{25}{100}$ (number of smokers)
- $P(Y = 1|X = 1) = \frac{P(X=1,Y=1)}{P(X=1)} = \frac{5/100}{25/100} = \frac{1}{5}$ (cancer if smoker)
- $P(Y = 1|X = 0) = \frac{P(X=0,Y=1)}{P(X=0)} = \frac{3/100}{75/100} = \frac{1}{25}$ (cancer if non smoker)
- $P(X = 1, Y = 1) \neq P(X = 1) \cdot P(Y = 1)$, thus X and Y are *not* independent of each other.

For continuous joint PDF $f_{X,Y}$:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx\,dy = 1$$

If X and Y are continuous with joint CDF $F_{X,Y}$, then their joint PDF is the derivative of the joint CDF $F_{X,Y}$ with respect to x and y. For joint PDF $f_{X,Y}$, the marginal PDF equals:

$$P(X = x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy$$

The conditional PDF is calculated the same as the conditional PMF.

2D LOTUS: Let g be a function from $\mathbb{R}^2$ to $\mathbb{R}$. If random variables X and Y are discrete:

$$E(g(X,Y)) = \sum_x \sum_y g(x,y)P(X = x, Y = y)$$

If X and Y are continuous with joint PDF $f_{X,Y}$:

$$E(g(X,Y)) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)f_{X,Y}(x,y)\,dx\,dy$$

The **covariance** between random variables X and Y is defined as:

$$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y)$$

This also means $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$. If $\mathrm{Cov}(X,Y) = 0$, then X and Y are **independent**. The **correlation** between X and Y is:

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}$$

$$\mathrm{Corr}(X,Y) \in [-1,1]$$

A k-dimensional vector $X = (X_1, ..., X_k)$ has a **multivariate normal distribution** if every linear combination of X has a normal distribution. This means $t_1X_1, ..., t_kX_k$ has a normal distribution for any constants $t_1, ..., t_k$.

Lecture 7: Moments

The **Median** of a random variable X is the exact midpoint of its probability.

$$P(X \leq c) \geq \frac{1}{2} \text{ and } P(X \geq c) \geq \frac{1}{2}$$

Half of the distribution falls to the left of c, the other half to the right of c. The means the CDF of X hits $\frac{1}{2}$ exactly at c (assuming it has no jumps).

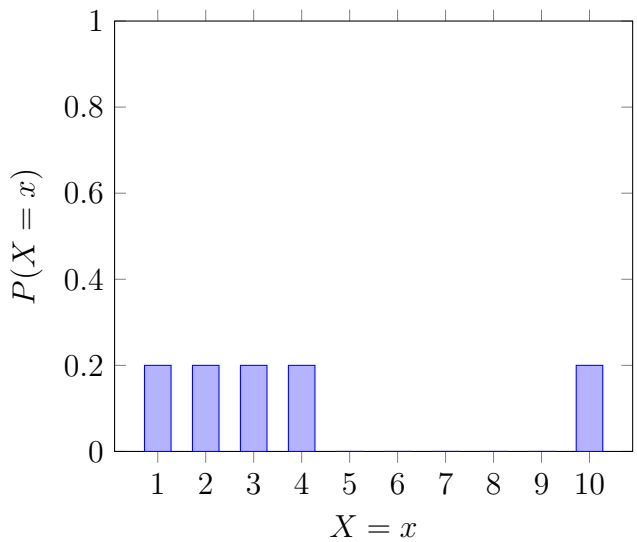


Figure 4: $E(X) = 4$, Median$(X) = 3$

Moments: Let X be a random variable with mean μ and variance σ^2:

- the nth **moment** of X is $E(X^n)$
- the nth **central moment** of X is $E((X - \mu)^n)$
- the nth **standardized moment** of X is $E((\frac{X-\mu}{\sigma})^n)$

These moments can also not exist. The **Moment Generating Function (MGF)** of a random variable is:

$$M(t) = E(e^{tX})$$

This MGF exists if $M(t)$ is finite on some open interval $(-a, a)$ containing 0. We can calculate the nth moment of X by evaluating the nth derivative of the MGF at 0:

$$E(X^n) = M^{(n)}(0)$$

If two random variables X and Y have the same MGF, they must have the same distribution. If X and Y are independent, then the MGF of $X + Y$ is the product of the individual MGFs:

$$M_{X+Y}(t) = M_X(t)M_Y(t)$$

Lecture 8: Inequalities and Limits

Markov's inequality states that for any random variable X and a constant $a > 0$:

$$P(|X| \geq a) \leq \frac{E|X|}{a}$$

Example: In a group of 100 people, is it possible that more than 50% are richer than at least twice the average (mean) person? This can be expressed as:

$$P(|X| \geq 2E(X)) \leq \frac{1}{2}$$

Chebyshev's inequality states that for any random variable X with mean μ and variance σ^2, and a constant $a > 0$:

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

For all positive integers n, let

$$\bar{X}_n = \frac{X_1 + ... + X_n}{n}$$

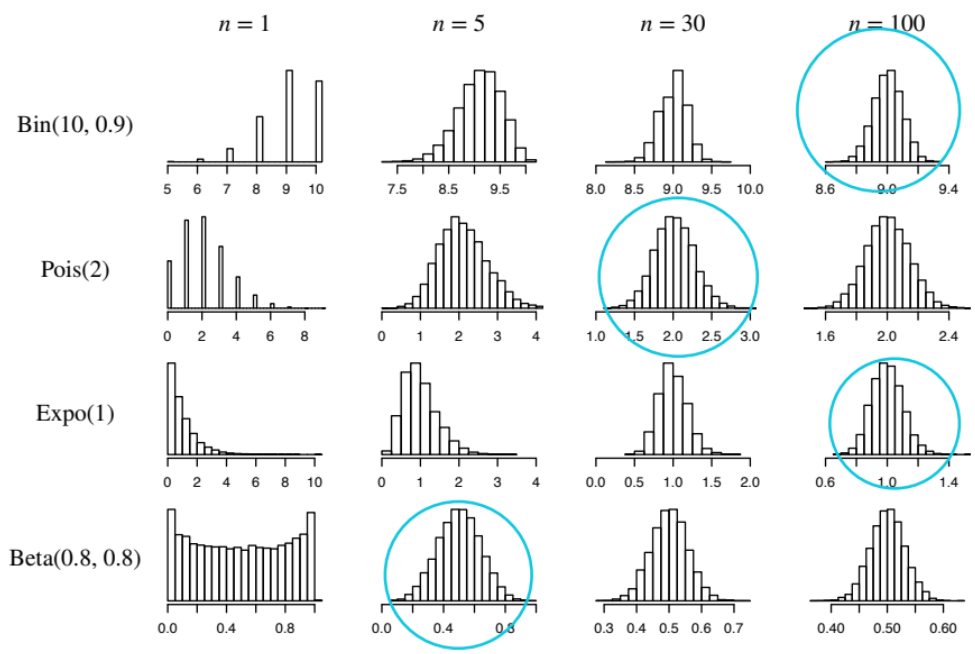
The **Weak Law of Large Numbers** states that for all $\epsilon > 0$:

$$P(|\bar{X}_n - \mu| \geq \epsilon) \to 0 \text{ as } n \to \infty$$

Central Limit Theorem:

$$\text{As } n \to \infty,\ \sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \to N(0,1) \text{ in distribution}$$

For large n, the distribution of $\bar{X}_n$ is approximately $N(\mu, \frac{\sigma^2}{n})$.



Lecture 9: Population and Samples

Population Mean:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Population Total:

$$\tau = \sum_{i=1}^{N} x_i = N\mu$$

Population Variance:

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu)^2$$

Sample surveys are used to obtain information about a large population by examining only a small fraction of that population. A **sample** is a subset of population N picked without replacement. We can estimate the population mean μ by calculating the **sampling mean** $\bar{X}_n$:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$$

The **sampling distribution** of $\bar{X}_n$ determines how accurately $\bar{X}_n$ estimates μ. For simple random sampling:

$$E(\bar{X}) = \mu$$

In general, if an estimator $\hat{\theta}$ of a population parameter θ is $E(\hat{\theta}) = \theta$, then $\hat{\theta}$ is **unbiased**.

In case of sampling *with* replacement (total independence):

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

With **standard error**:

$$\sigma_{\bar{X}_n} = \frac{\sigma}{\sqrt{n}}$$

In case of sampling *without* replacement (some dependences):

$$\mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}\left(1 - \frac{n-1}{N-1}\right) \approx \frac{\sigma^2}{n}$$

With **standard error**:

$$\sigma_{\bar{X}_n} = \frac{\sigma}{\sqrt{n}}\sqrt{1 - \frac{n-1}{N-1}}$$

If n is large, but still small relative to N, then $\bar{X}_n$ is approximately normal distributed.

Confidence Interval: For what upper bounds u and v is the probability of random variable X a certain $1 - \alpha$:

$$P(u \leq X \leq v) = 1 - \alpha$$

$$u, v = \bar{X}_n \pm z(\alpha/2) \cdot \sigma_{\bar{X}_n}$$

For the following z-values:

α	z
0.10	1.65
0.05	1.96
0.01	2.58

Lecture 10: Parameter Estimation

Sample Moment:

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k$$

$\hat{\mu}_k$ is an estimate of the actual kth moment $\mu_k = E(X^k)$.

The **likelihood** of parameter θ is the probability of given data $x_1, x_2, ..., x_n$, that θ is a parameter of this data:

$$\text{lik}(\theta) = f(x_1, x_2, ..., x_n|\theta)$$

If the X_i are independent and identically distributed, then:

$$\text{lik}(\theta) = \prod_{i=1}^{n} f(x_i|\theta)$$

The **Maximum Likelihood Estimate** equals the maximum of:

$$\sum_{i=1}^{n} \log\left[f(x_i|\theta)\right]$$

Consistent estimate: $\hat{\theta}$ converges to θ as $n \to \infty$.

Example of MLE:
Let $X_1, X_2, ..., X_m$ be independent and identically distributed Binomial random variables. The PMF of a Binomial random variable is:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

with number of trials n and success probability p. Find the MLE of p (both n and m are constants).

Solution: Since $\binom{n}{k}$ is not a function of p we can ignore it. We can use the Bernoulli random variables $Y_1, Y_2, ..., Y_{n\times m}$ (the n from the Binomial is integrated into the $n \times m$, which is why we can use Bernoulli variables). The likelihood of these Bernoulli variables is:

$$L_{n\times m}(p) = \prod_{i=1}^{n\times m} p^{Y_i}(1-p)^{1-Y_i}$$

Multiplying this out we get:

$$L_{n\times m}(p) = p^S(1-p)^{n\times m-S} \text{ where } S = \sum_{i=1}^{n\times m} Y_i$$

The log-likelihood:

$$= \log\left(p^S(1-p)^{n\times m-S}\right)$$

$$= \log\left(p^S\right) + \log\left((1-p)^{n\times m-S}\right)$$

$$= S\log(p) + (n\times m - S)\log(1-p)$$

Take the derivative and set to 0:

$$l'_{n\times m}(p) = \frac{S}{p} - \frac{n\times m - S}{1-p} = 0$$

Solving for p gives:

$$p = \frac{S}{n\times m}$$

Lecture 11: Hypothesis Testing

Imagine you have 2 coins; one with a probability of 0.5 and one with a probability of 0.7. The probabilities for throwing $X = x$ number of heads looks like this:

$X = x$	0	1	2	3	4	5	6	7	8	9	10
H_0	.001	.0098	.0439	.1172	.2051	.2461	.2051	.1172	.0435	.0058	.001
H_1	.000	.0001	.0014	.0090	.0368	.1029	.2001	.2668	.2335	.1211	.0282
Ratio	100	98	31.36	13.02	5.57	2.39	1.02	.439	.186	.048	.035

Here H_0 represents the **null hypothesis** (that the coin with 0.5 probability is being used), and H_1 represents the **alternative hypothesis** (that the coin with 0.7 probability is being used). We can specify a **threshold** c for when we accept or reject the null hypothesis based on the **ratio** of the probabilities. For example, if we specify a threshold of $c = 1$ then we **accept** H_0 if $X \leq 6$, and we **reject** H_0 if $X > 6$:

$$P(\text{reject } H_0|H_0) = P(X > 6|H_0) = 0.1675 \text{ (sum fields of } H_0)$$

$$P(\text{accept } H_0|H_1) = P(X \leq 6|H_1) = 0.3503 \text{ (sum fields of } H_1)$$

For $c = 0.1$ we **accept** H_0 if $X \leq 8$, and we **reject** H_0 if $X > 8$:

$$P(\text{reject } H_0|H_0) = P(X > 8|H_0) = 0.0068 \text{ (sum fields of } H_0)$$

$$P(\text{accept } H_0|H_1) = P(X \leq 8|H_1) = 0.8506 \text{ (sum fields of } H_1)$$

Some terminology:

- **Type I error:** rejecting H_0 when it is true
- **Significance Level α:** $P(\text{reject } H_0|H_0)$
- **Type II error:** accepting H_0 when it is false
- β**:** $P(\text{accept } H_0|H_1)$
- **Power** $1-\beta$**:** $P(\text{reject } H_0|H_1)$
- **Rejection Region:** set of values $X = x$ that lead to rejection of H_0
- **Acceptance Region:** set of values $X = x$ that lead to accepting H_0
- **Null Distribution:** distribution of X when H_0 is true

P-value: the smallest α for which H_0 is rejected.
e.g. given $X = 7$, the p-value equals $P(X \geq 7|H_0)$.